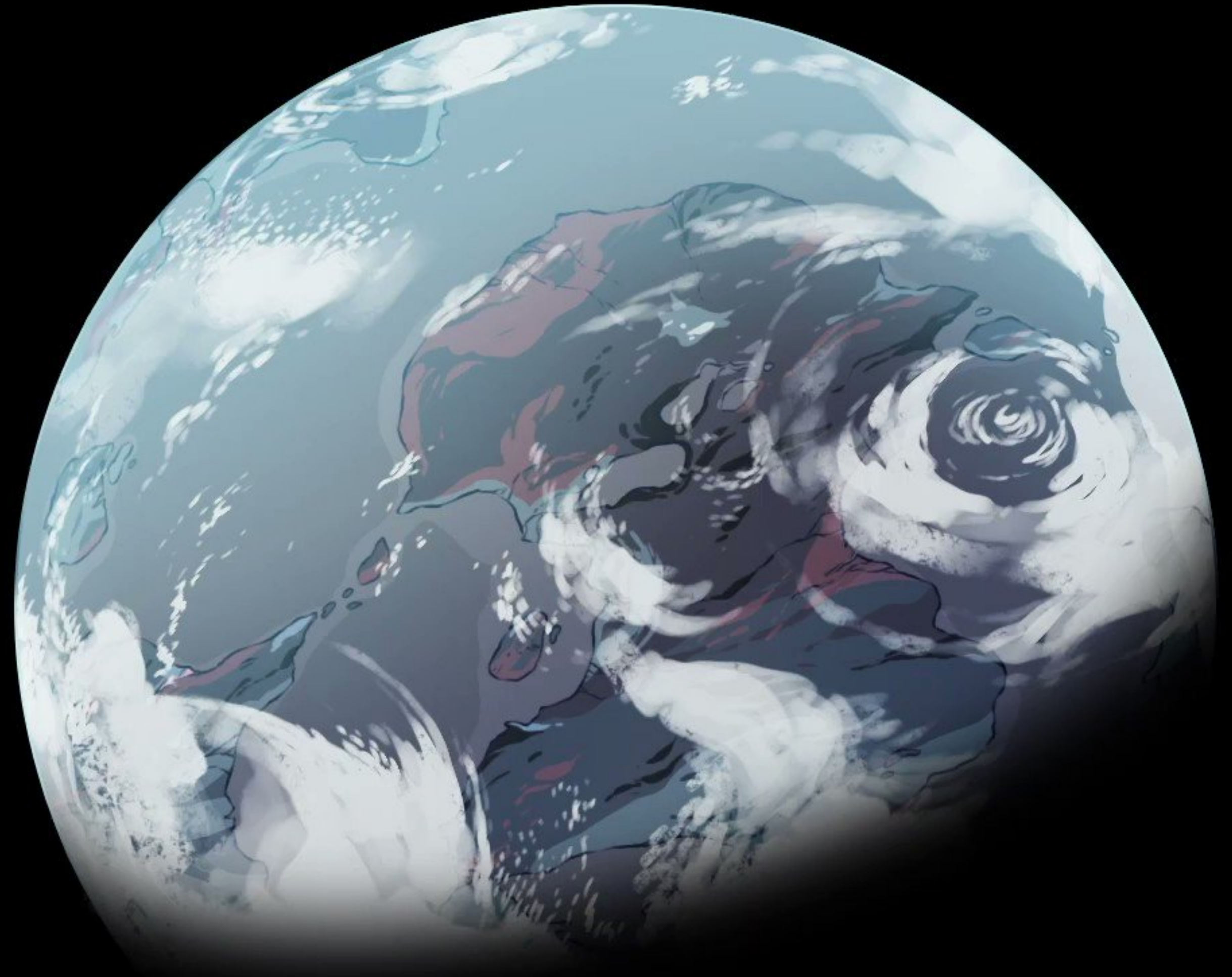


Science & Science Fiction

33-120 | Prof. Katie Breivik



Announcements

- ▶ Midterm Film Critique due next Monday 02/24
- ▶ Problem 4 is available and based on today's lecture
 - ▶ Due at midnight on Wednesday 02/26
- ▶ Exploration paper 2 due next Friday 02/28

Today: Optical properties of solids, Cloaking

optical properties of solids

scattering

rough surfaces
or small
particles

Air molecules scatter
light; blue light with
shortest λ gets
scattered the most

reflection

smooth
surfaces

mirrors

transmission

transparent
materials

material doesn't
interact with the
light; windows

absorption

Non-
transparent
materials

material *does*
interact with the
light; colors!

scattering



transmission



reflection



absorption





Star Trek: The Voyage Home

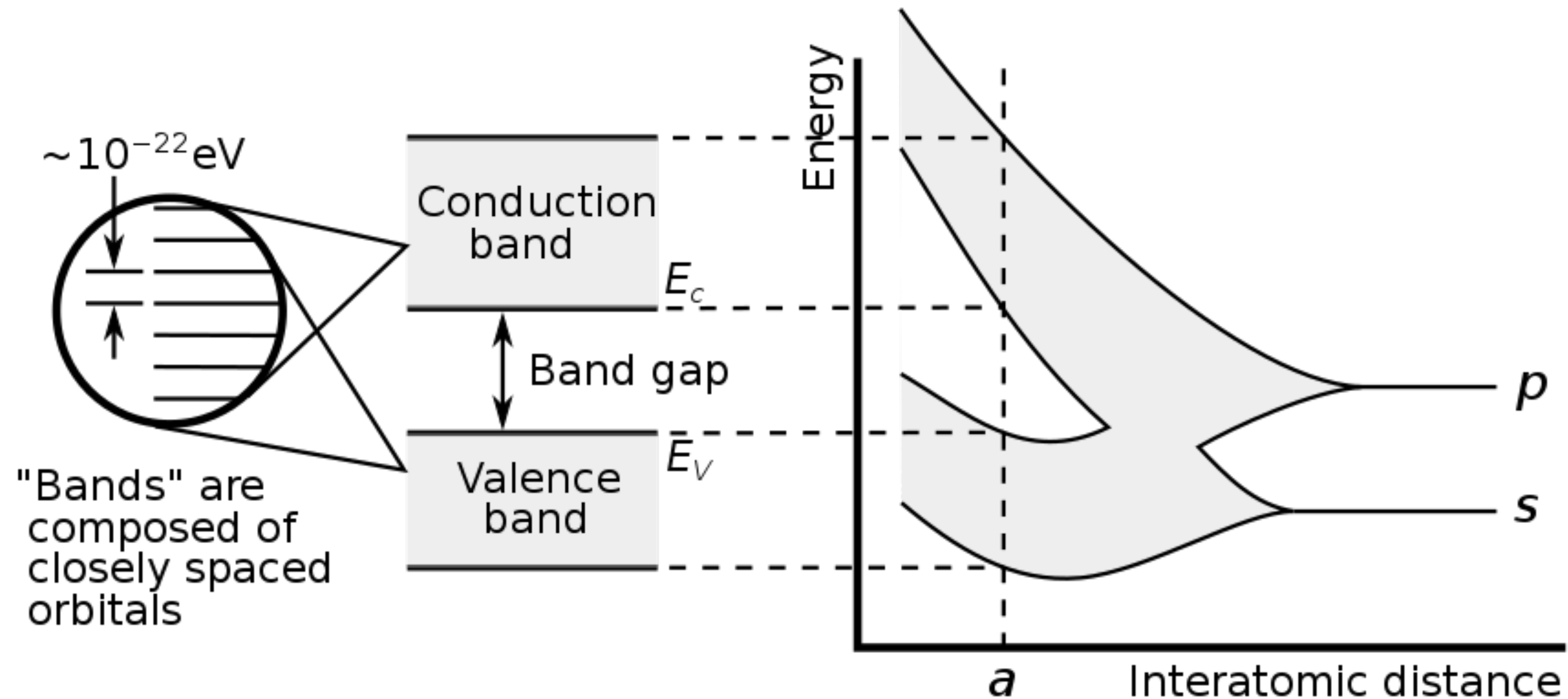
Directed:
Leonard Nimoy

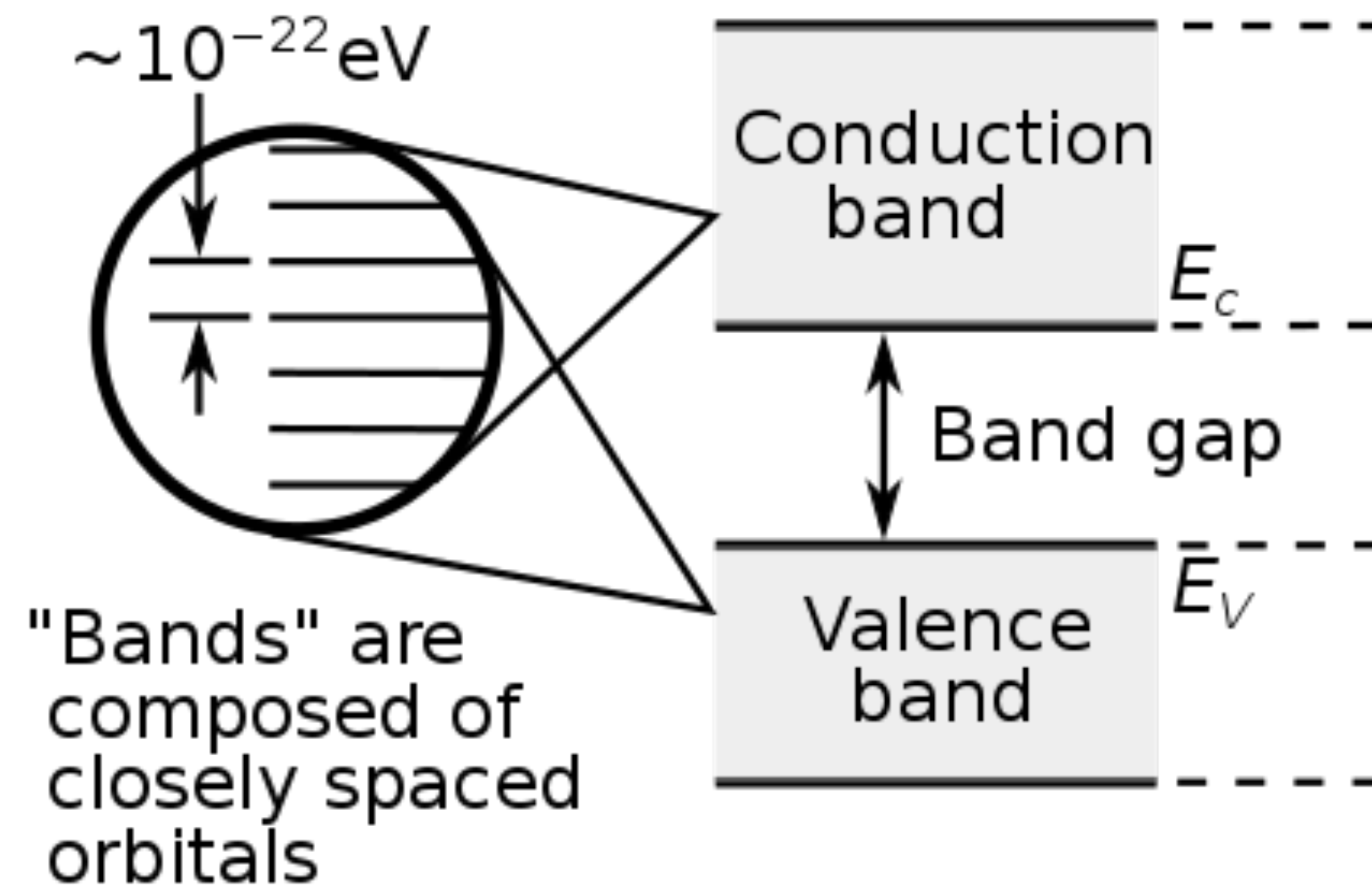
United Artists
(1986)



What is
“transparent
aluminum”?

optical properties of solids depend on
the electronic band structure of the material

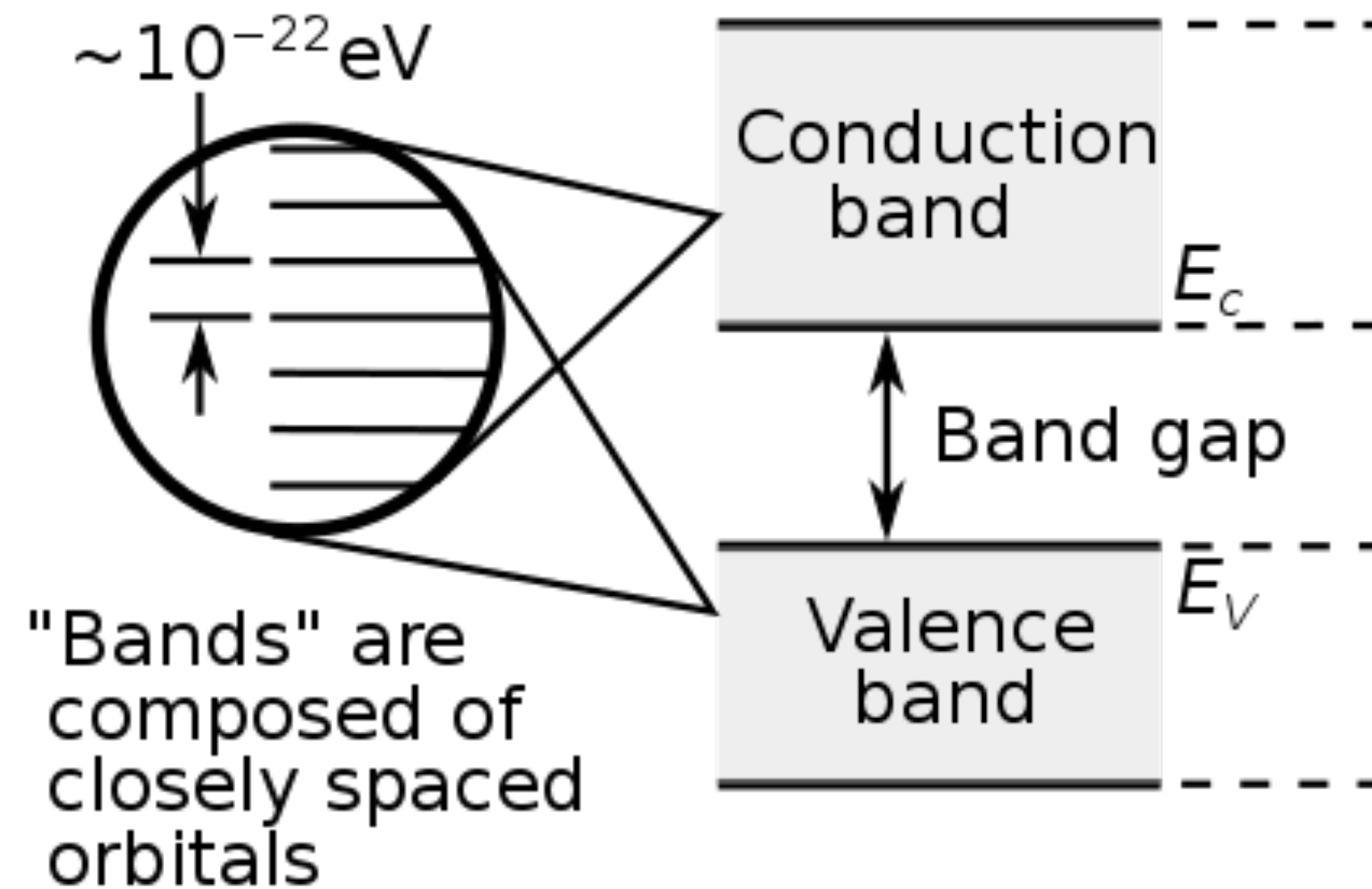




Conduction band: electrons are free to move throughout the band

Band gap: electrons are forbidden from having energies in this range

Valence band: electrons are shared between neighboring atoms; move with **quantum tunneling**



Conduction band: electrons are free to move throughout the band

Band gap: electrons are forbidden from having energies in this range

quantum tunneling

electrons pass through energy barriers with an extremely low, but nonzero, probability

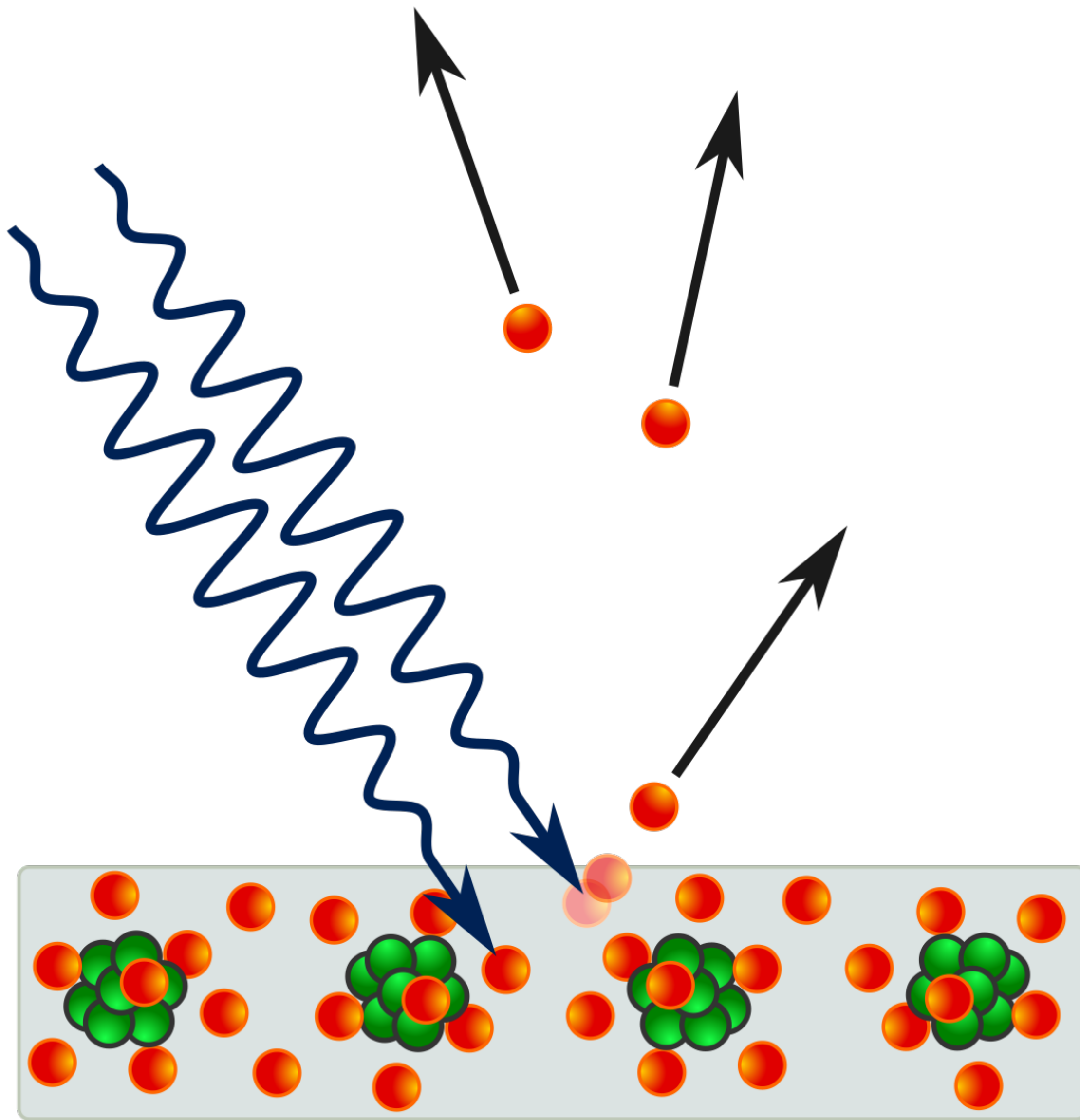
Valence band: electrons are shared between neighboring atoms; move with **quantum tunneling**

photoelectric effect

photons with the correct energy are absorbed by the material and eject conduction or valence electrons

(a big year!)

Described by Einstein in 1905
and was awarded Nobel prize
in 1921

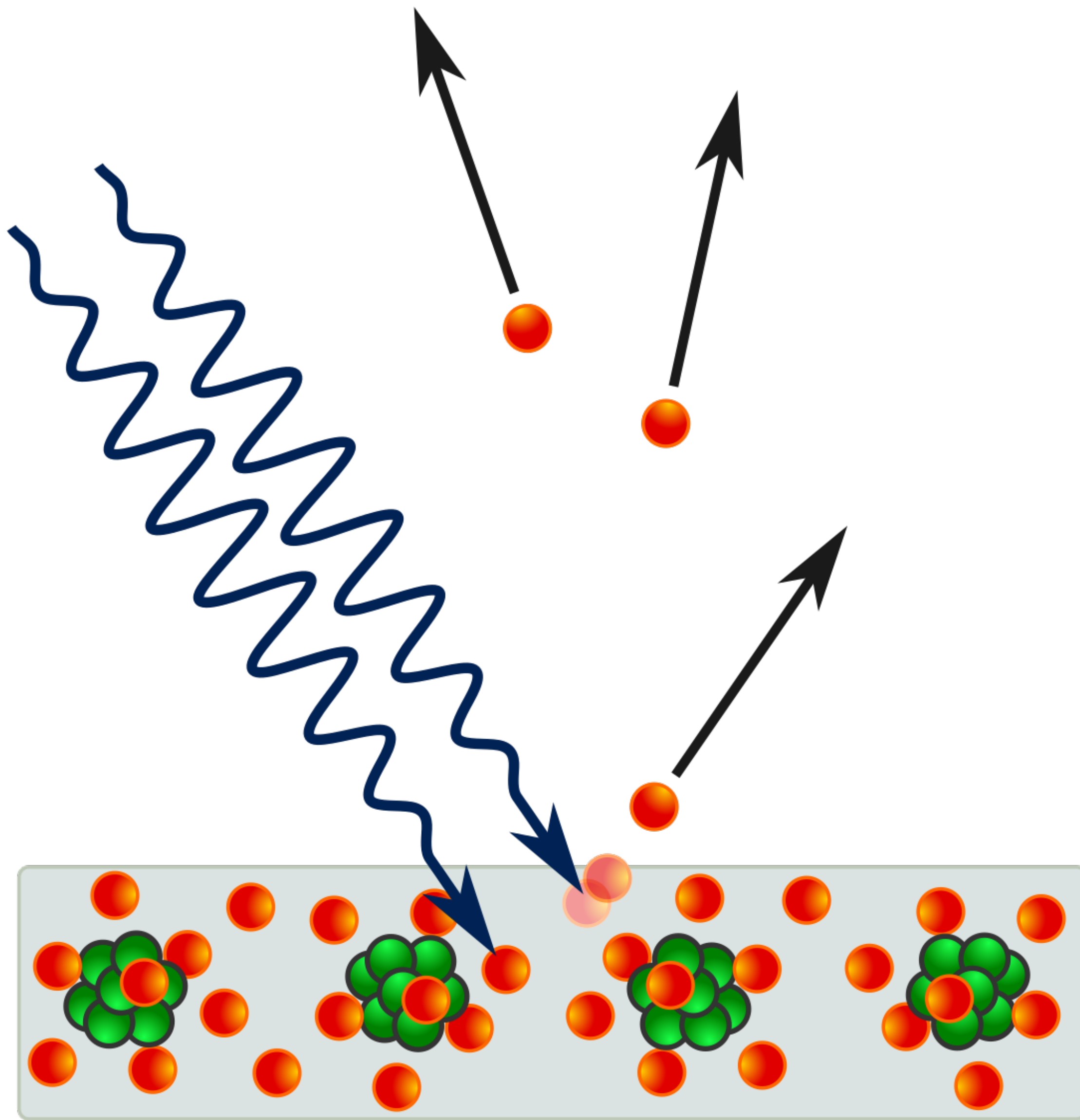


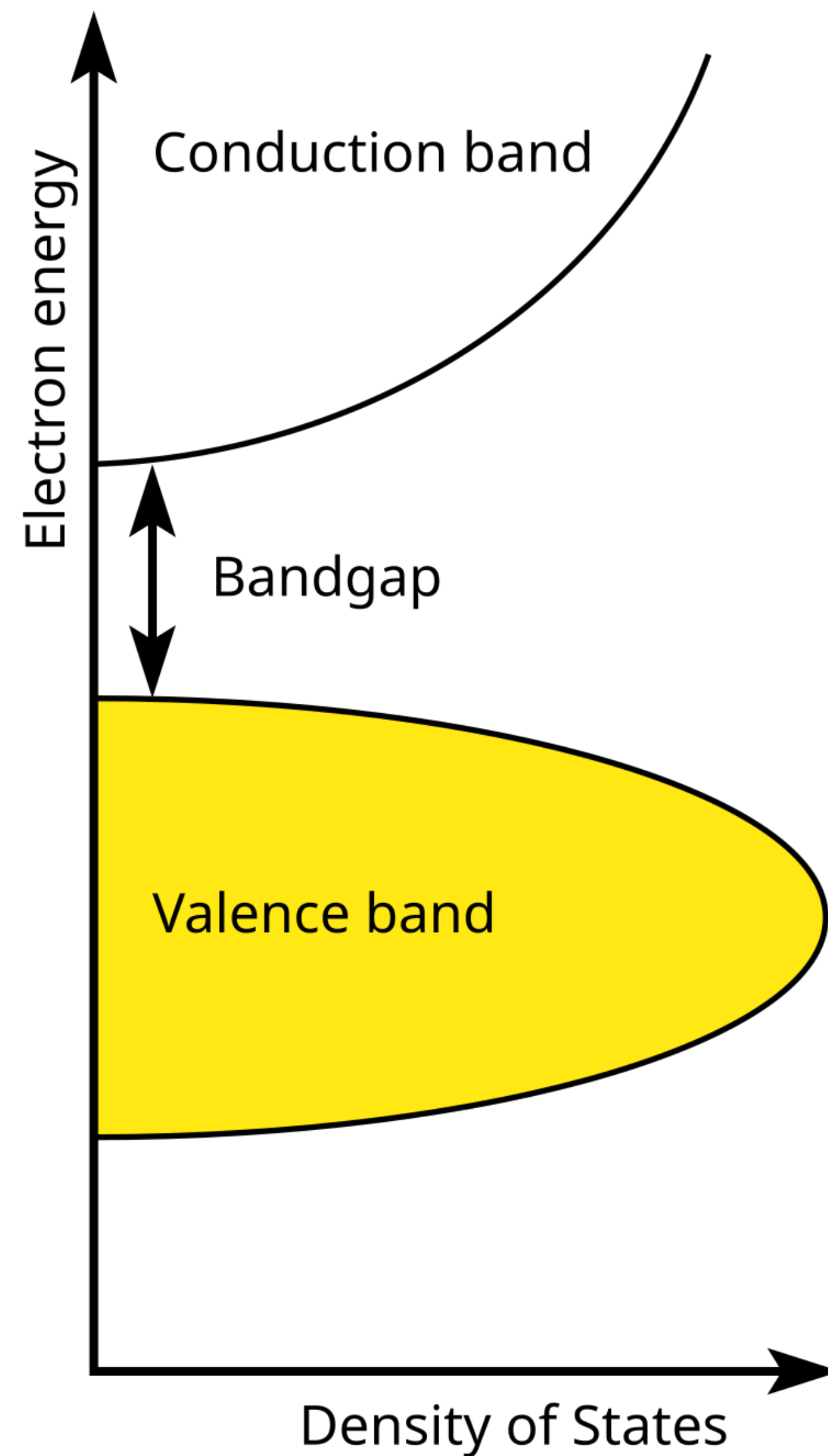
photoelectric effect

photons with the correct energy are absorbed by the material and eject conduction or valence electrons

Different photon energies can eject electrons in different conduction and valence bands

The material content sets the energy





band gaps

Photons are *not* absorbed in band gaps

The larger the band gap, the wider the energy range of light that passes through

conduction/ valence bands

Photons are absorbed in the bands

Conductors:

Overlapping conduction & valence bands

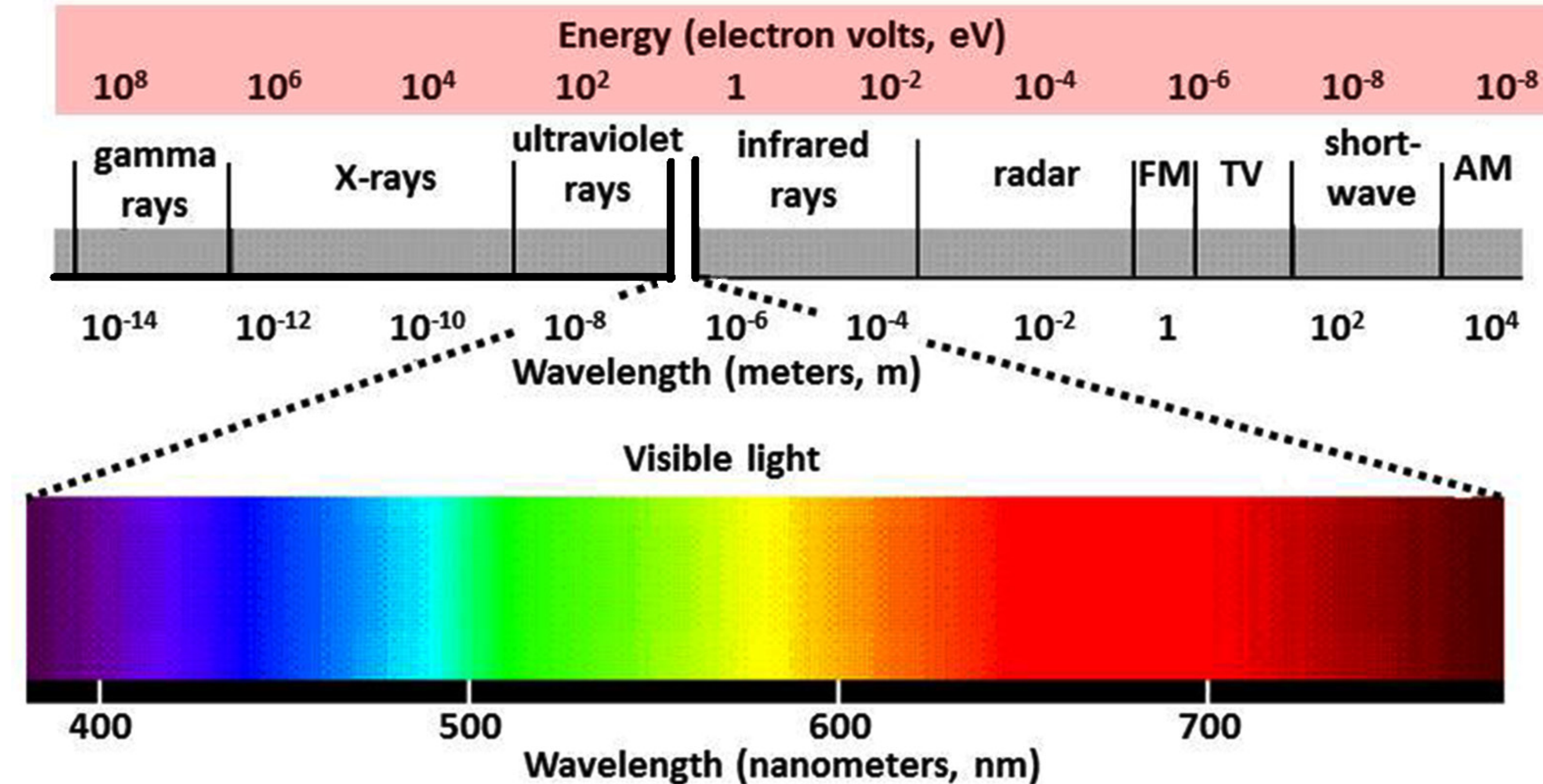
Semiconductors:

Small bandgap between conduction & valence bands

Insulators:

Large bandgap between conduction & valence bands

Materials can be transparent or opaque to all wavelengths of light



Small bandgap materials

➤ Aluminum	0	(metal)	} conductors
➤ Graphite	0	(semimetal)	
➤ Germanium	0.67		} semiconductors
➤ Silicon	1.1		

Visible photons: $E \sim 2\text{-}3\text{ eV}$

These materials are **opaque** to visible light

Large bandgap materials

➤ Diamond	5.5	} insulators
➤ SiO₂	9	

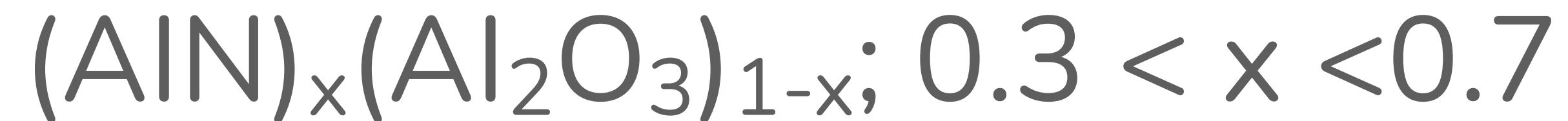
Visible photons: $E \sim 2\text{-}3\text{ eV}$

These materials are **transparent** to visible light

Back to transparent aluminum

Is there an insulator type of aluminum?

Aluminum Oxynitride (ALON)



Patented in 1985; the hardest transparent polycrystalline material available —> bullet-proof windows

From transparency to *invisibility*

Are invisibility cloaks
physically possible?

What is necessary to render
something totally invisible?

How does light respond to striking a surface?

- ▶ Reflection: bounces off of smooth surface or scatters off of rough surface
- ▶ Refraction: bends as it passes through a transparent object
- ▶ Absorption: energy is transmitted to the surface/material

3 common tricks to simulate invisibility

1. Camouflage: cover with something that blends into the background
2. Stealth technology: cover with something that absorbs radar & modify shape to minimize reflection
3. Cloaking: bend light around the object



Die Another Day

Directed:
Lee Tamahori

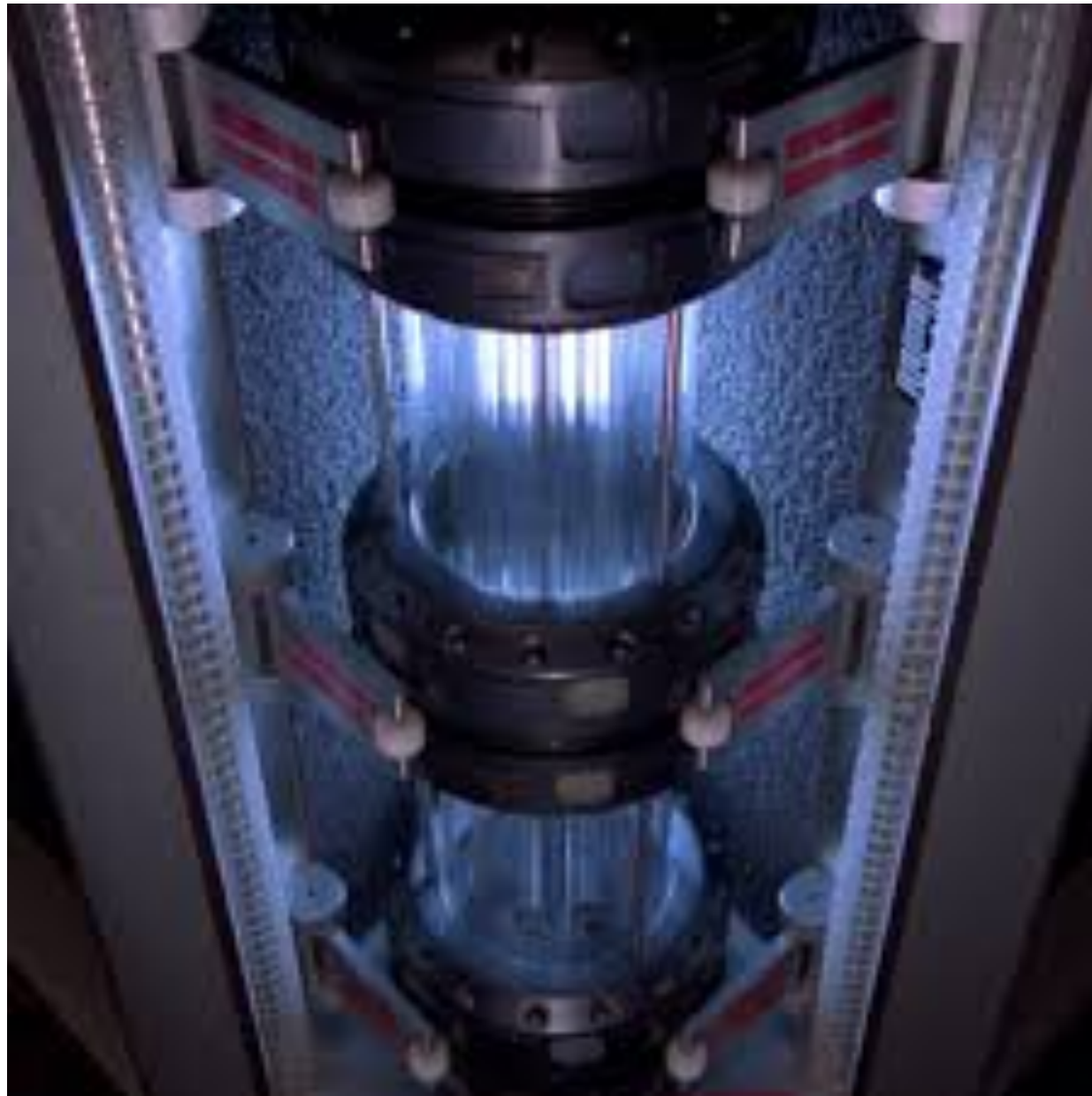
MGM (2002)

1. Camouflage: cover with something that blends into the background



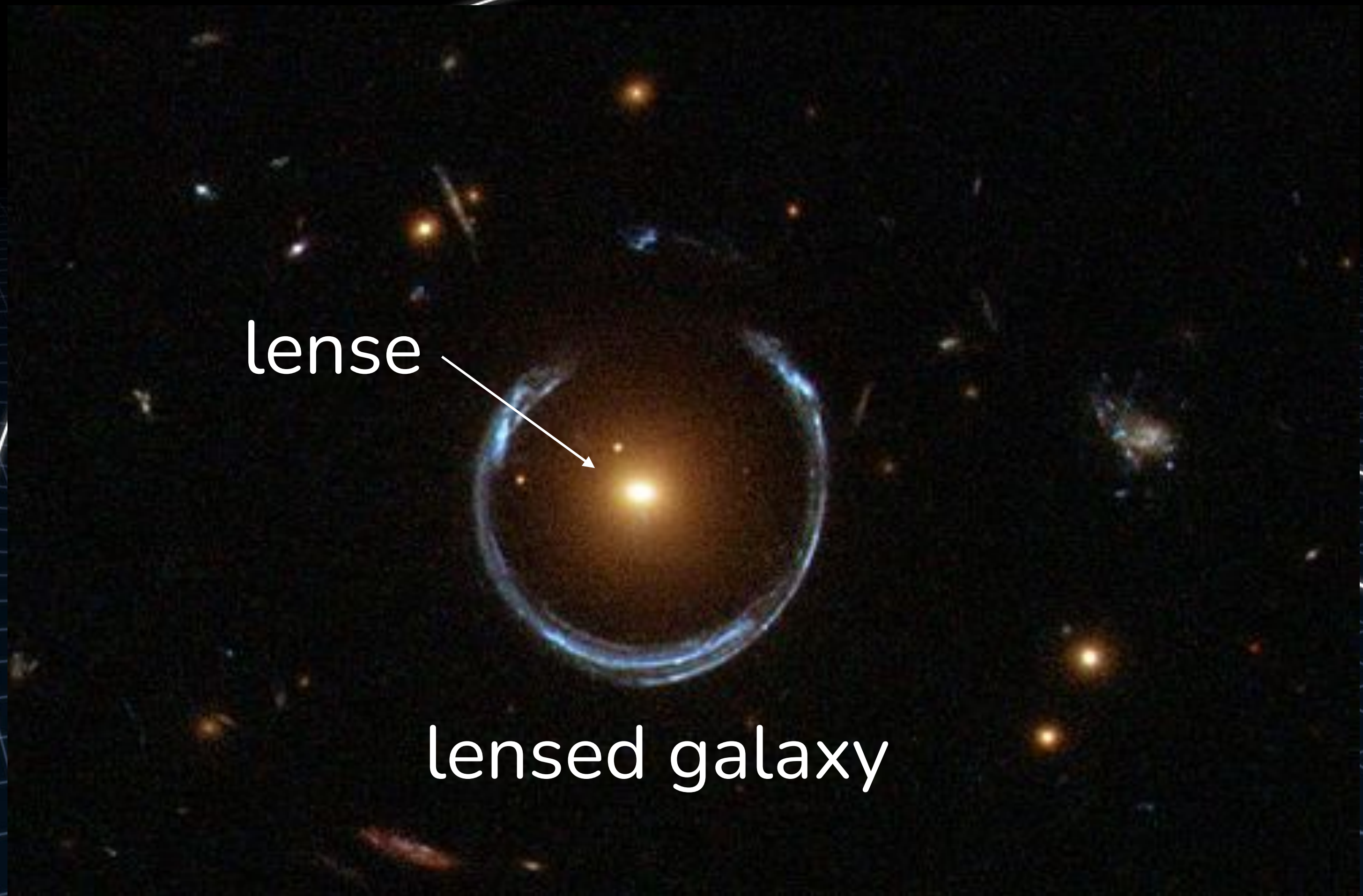
2. Stealth technology: cover with something that absorbs radar & modify shape to minimize reflection — works best at night

Applied in multiple sci-fi contexts: in Independence Day
B-2 bombers are deployed against alien attack ships



3. Cloaking: bend light around the object

Used in many different iterations in the star trek universe by the Romulans, Klingons and sometimes the United Federation of Planets



Gravitational lensing is a **real physical concept** used to both measure the mass of the lens (based on how much the light curves) and see objects that would be too dim without the lens



Harry Potter and the Sorcerer's Stone

Directed:
Chris Columbus

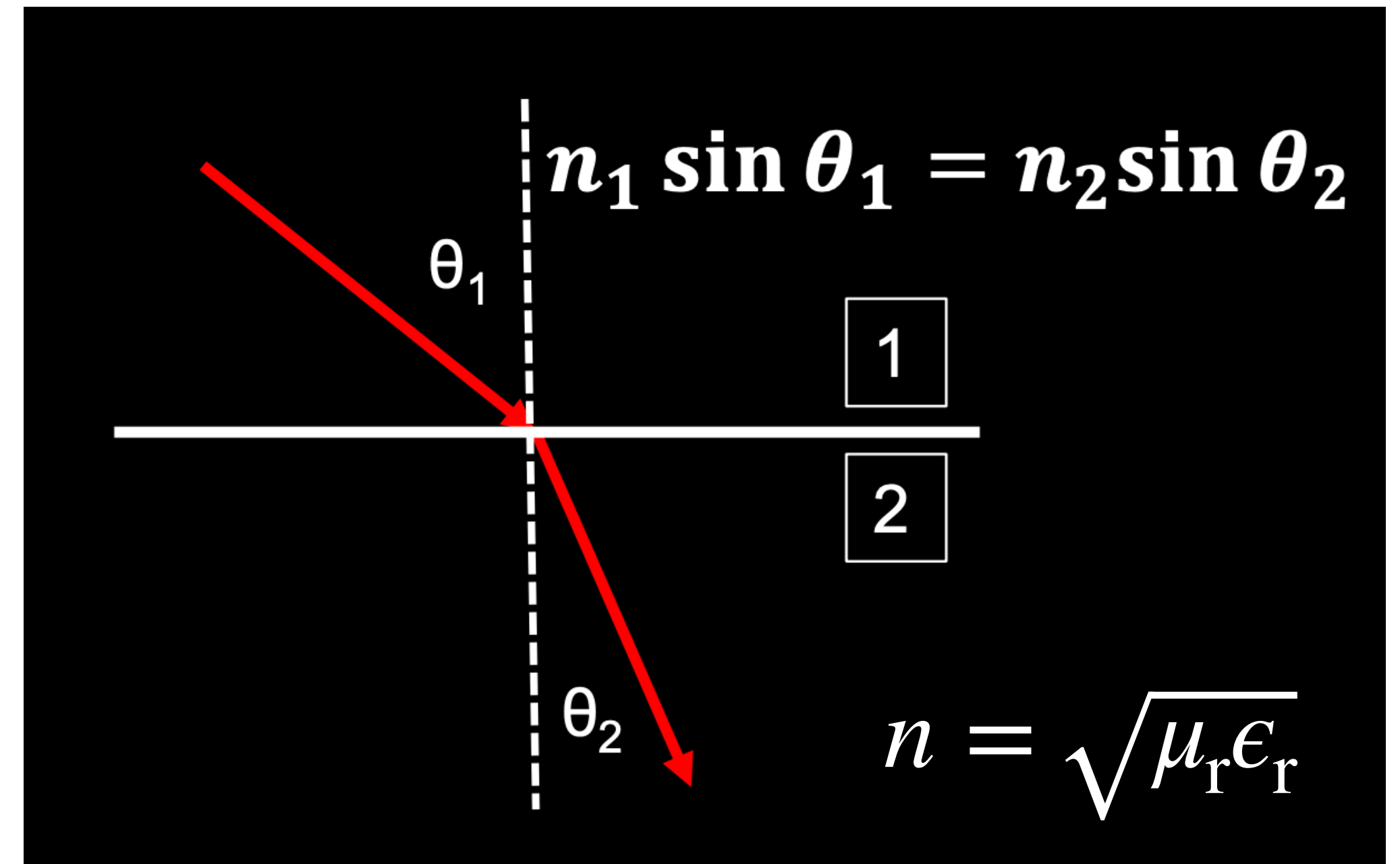
Warner Brothers
(2001)

Is this a secret, fourth method to simulate invisibility?

Is this a secret, fourth method to simulate invisibility?

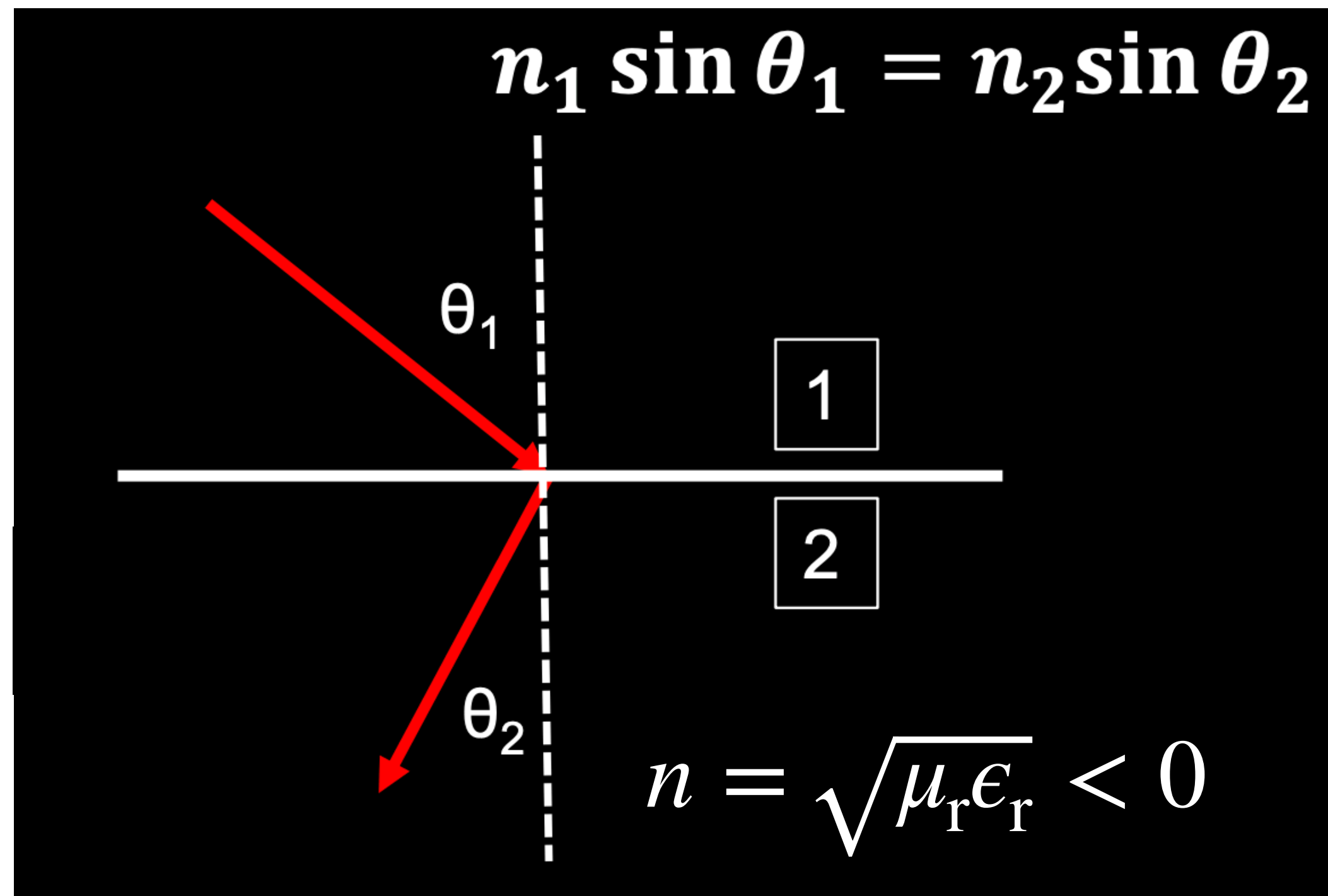
Light interacts with *ordinary materials* through

- ▶ Reflection: bounces off of smooth surface or scatters off of rough surface
- ▶ Refraction: bends as it passes through a transparent object
- ▶ Absorption: energy is transmitted to the surface/material



The index of refraction, n , tells us how light bends when crossing surface into a new material — depends on electric (ϵ_r) & magnetic (μ_r) properties of the material

Is this a secret, fourth method to simulate invisibility?

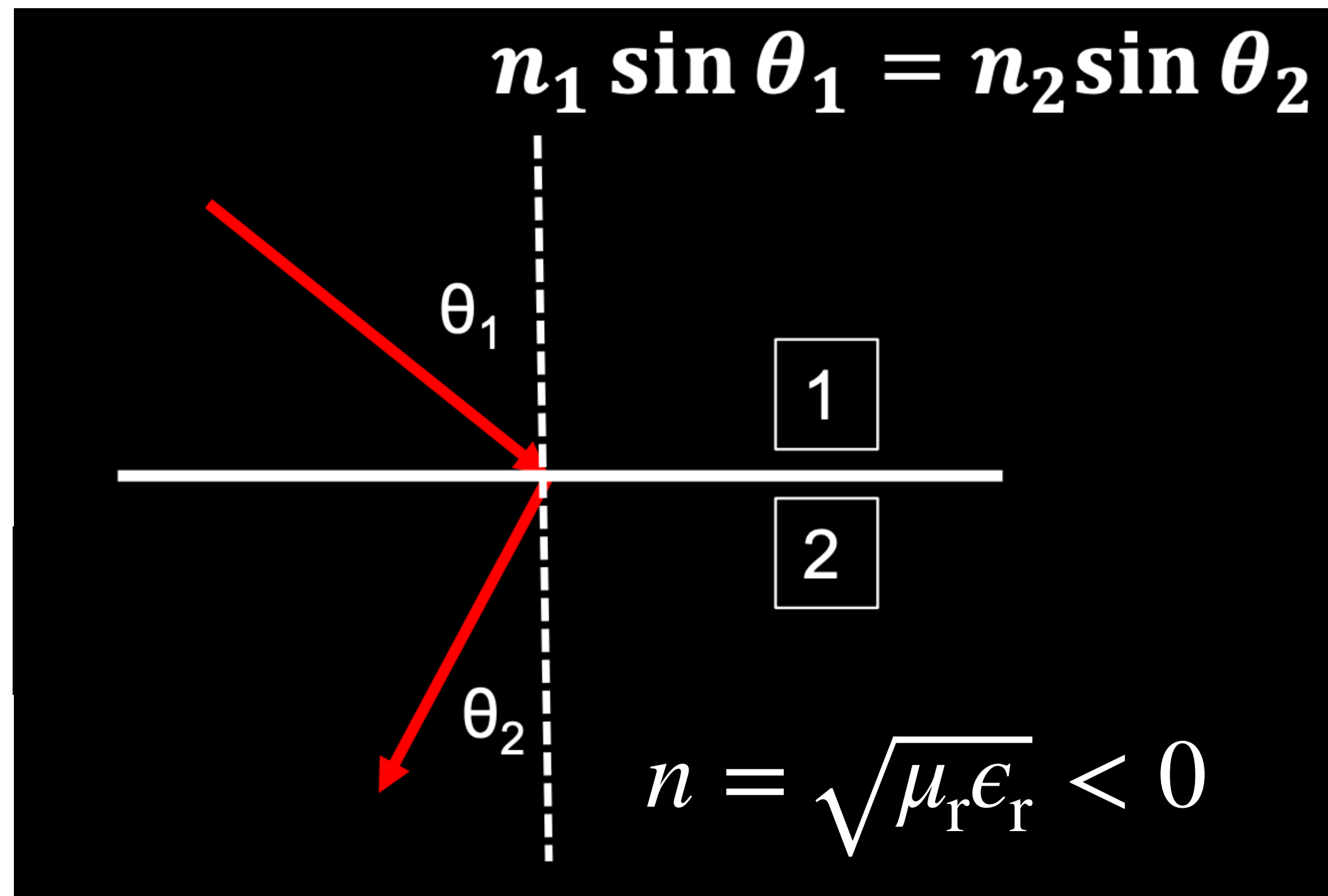


$n < 0 \rightarrow$ light is reflected
away from observer

Light experiences refraction
with *metamaterials*
differently than ordinary
materials

All indices of refraction have a
real and imaginary part; the
real part sets the reflection and
refraction & the imaginary part
describes absorption

Is this a secret, fourth method to simulate invisibility?



$n < 0 \rightarrow$ light is reflected
away from observer

Light experiences refraction
with *metamaterials*
differently than ordinary
materials

First shown with microwaves in
2006; so far has only been
shown for other narrow bands
across the electromagnetic
spectrum

Reminders!

- ▶ Midterm Film Critique due next Monday 02/24
- ▶ Problem 4 due next Wednesday 02/26
- ▶ Exploration paper 2 due next Friday 02/28

Next time: Energy & Power